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Bayesian estimation of genomic clines

ZACHARIAH GOMPERT and C. ALEX BUERKLE

Department of Botany and Program in Ecology, University of Wyoming, Laramie, WY 82071, USA

Abstract

We developed a Bayesian genomic cline model to study the genetic architecture of adaptive divergence and reproductive isolation between hybridizing lineages. This model quantifies locus-specific patterns of introgression with two cline parameters that describe the probability of locus-specific ancestry as a function of genome-wide admixture. 'Outlier' loci with extreme patterns of introgression relative to most of the genome can be identified. These loci are potentially associated with adaptive divergence or reproductive isolation. We simulated genetic data for admixed populations that included neutral introgression, as well as loci that were subject to directional, epistatic or underdominant selection, and analysed these data using the Bayesian genomic cline model. Under many demographic conditions, underdominance or directional selection had detectable and predictable effects on cline parameters, and 'outlier' loci were greatly enriched for genetic regions affected by selection. We also analysed previously published genetic data from two transects through a hybrid zone between *Mus domesticus* and *M. musculus*. We found considerable variation in rates of introgression across the genome and particularly low rates of introgression for two X-linked markers. There were similarities and differences in patterns of introgression between the two transects, which likely reflects a combination of stochastic variability because of genetic drift and geographic variation in the genetic architecture of reproductive isolation. By providing a robust framework to quantify and compare patterns of introgression among genetic regions and populations, the Bayesian genomic cline model will advance our understanding of the genetics of reproductive isolation and the speciation process.

Keywords: admixture, Bayesian inference, hybrid zones, introgression, reproductive isolation, speciation

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Introduction

Natural hybridization and introgression provide a unique opportunity to study speciation and the genetics of reproductive isolation (Endler 1977; Hewitt 1988; Barton & Hewitt 1989; Arnold 1997; Buerkle & Lexer 2008). Empirical data indicate that hybridization is common in nature and that different genetic regions are exchanged between hybridizing lineages at different rates (e.g. Rieseberg *et al.* 1999; Lexer *et al.* 2007; Macholan *et al.* 2007; Carling & Brumfield 2009; Nolte *et al.* 2009; Gompert *et al.* 2010; Teeter *et al.* 2010). These observations demonstrate that at the genomic level species boundaries are not absolute and monolithic, but rather that a subset of genetic regions contribute to reproductive isolation and resist introgression (Hewitt 1988; Wu 2001; Mallet 2005). This occurs because independent assortment and recombination in admixed individuals can mix parental genotype combinations and break up blocks of parental ancestry (Barton & Hewitt 1985; Buerkle & Lexer 2008). The rate and form of introgression for each genetic region is shaped by the effect of its allelic variation on fitness and linkage disequilibrium with other genetic regions. In natural hybrid zones, this includes environment-independent and environment-dependent fitness effects, as well as variable fitness because of epistatic interactions and genomic background. Genetic regions with allelic variants that reduce viability or fertility in hybrids or contribute to

Correspondence: Zachariah Gompert, Fax: (307) 766 2851;
E-mail: zgompert@uwyo.edu

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assortative mating are expected to experience limited introgression into alternative genomic backgrounds (Barton 1979; Rieseberg *et al.* 1999; Mallet 2005; Turner *et al.* 2005). Conversely, genetic regions harbouring an allele with high marginal fitness should undergo rapid introgression with the favoured allele spreading into the alternative genomic background (adaptive introgression; Barton 1983; Whitney *et al.* 2006; Fitzpatrick *et al.* 2009; Nolte *et al.* 2009). Expectations for the rate of introgression of neutral alleles are complex and depend on patterns of linkage disequilibrium, but should generally exceed rates of introgression for genetic regions associated with reproductive isolation (Barton 1983; Barton & Hewitt 1985; Turner *et al.* 2005). In general, linkage should cause correlated patterns of introgression for syntenic genetic regions, although the size of these correlated chromosomal blocks is a function of recombination and the number of generations of admixture (Buerkle & Rieseberg 2008). Quantifying patterns of differential introgression in admixed populations can provide insights into the speciation process and the genetic architecture of reproductive isolation between divergent lineages.

Patterns of introgression in natural hybrid zones or admixed populations have been quantified using the relationship between geographic location along a transect and population allele frequencies (geographic cline analysis; Barton & Hewitt 1985; Szymura & Barton 1986; Payseur *et al.* 2004; Yanchukov *et al.* 2006) or based on the frequency of locus-specific genotypes along a genome-wide admixture gradient (genomic cline analysis; Szymura & Barton 1986; Rieseberg *et al.* 1999; Lexer *et al.* 2007; Gompert & Buerkle 2009; Nolte *et al.* 2009). Genetic regions potentially associated with reproductive isolation can be inferred based on contrasting widths of geographic clines or discordance of genomic clines with a null model (Rieseberg *et al.* 1999; Payseur *et al.* 2004; Carling & Brumfield 2009; Gompert & Buerkle 2009; Teeter *et al.* 2010). Several empirical patterns have begun to emerge from hybrid zone research. Many studies have found variation among genetic regions in rates of introgression and substantially reduced rates of introgression for loci on the X chromosome (Payseur *et al.* 2004; Macholan *et al.* 2007; Teeter *et al.* 2010). Studies have also documented both similarities and differences in patterns of introgression across replicate hybrid zones (Szymura & Barton 1991; Buerkle & Rieseberg 2001; Yanchukov *et al.* 2006; Nolte *et al.* 2009; Teeter *et al.* 2010) and considerable variation in the number of genes likely contributing to reproductive isolation between hybridizing lineages (Szymura & Barton 1991; Porter *et al.* 1997; Macholan *et al.* 2007).

Geographic and genomic cline analysis are powerful methods that have led to important insights into the speciation process, but these methods also have several limitations. Geographic cline models assume a monotonic change in allele frequency along a linear geographic transect, which is not always a reasonable assumption particularly in patchy or mosaic hybrid zones (Harrison & Rand 1989; Vines *et al.* 2003; Payseur 2010). Also, there is not a clear optimal scale for geographic cline analysis, and the scale of sampling has the potential to affect results. Although genomic cline analysis circumvents some of these spatial issues, current genomic cline models (Gompert & Buerkle 2009) also suffer several limitations. Most importantly, it is difficult to derive an appropriate null distribution for genomic clines. Previous genomic cline models either require fixed differences between parental populations for the studied loci (permutation-based null model) or assume that genetic drift has not occurred in the admixed population (parametric null model; Gompert & Buerkle 2009). The latter assumption may be approximately true when admixture is recent, the admixed population is quite large, or the admixed population experiences high rates of gene flow with the parental populations, but will not hold in many cases (Fitzpatrick *et al.* 2009; Gompert & Buerkle 2009). Moreover, previous genomic cline models do not account for population structure within a hybrid zone or among admixed populations (although comparison of clines among admixed populations is possible using constrained models and likelihood ratio tests; Gompert & Buerkle 2009).

In this study, we develop a hierarchical Bayesian model for estimating genomic clines (i.e. clines in locus-specific ancestry along a genome-wide admixture gradient). This Bayesian model overcomes many of the limitations of earlier genomic cline models (Lexer *et al.* 2007; Gompert & Buerkle 2009) including the following. Our use of a Bayesian model provides robust inference of genomic cline parameters and uncertainty in these parameter estimates. This model does not require fixed allelic differences between parental populations. As this is a hierarchical model, it is possible to derive a null, genome-level distribution for genomic cline parameters directly from the data without assuming a specific demographic model for the admixed population (e.g. Guo *et al.* 2009; Gompert *et al.* 2010a). If most genetic regions do not affect fitness in admixed individuals, this null distribution should approximate neutral expectations for rates of introgression. Moreover, even if allelic variants throughout much of the genome affect fitness, the most extreme genomic clines should still correspond to many of the genetic regions most associated with reproductive isolation. The proposed genomic cline model also provides a means to model similarities and differences in patterns of introgression among admixed

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populations, including well-sampled demes along a transect through a single hybrid zone. Despite these advances, the proposed model retains two important limitations of the likelihood-based genomic cline model: it does not account for tight linkage among genetic markers, and the null distribution cannot be precisely equated with a neutral distribution (for details see Discussion).

We begin by describing the hierarchical Bayesian genomic cline model. We then use simulated data to explore expected variation among genetic regions in genomic clines under models of neutrality. We explore additional simulated data that include selection and ask whether loci with extreme or 'outlier' genomic clines are enriched for genetic regions that were targets of selection. Finally we apply the Bayesian genomic cline model to previously published genetic data from two transects through a hybrid zone between house mice (*Mus domesticus* and *M. musculus*).

Methods

We define a Bayesian model for estimating genomic clines, with the primary aim of estimating cline parameters that describe the pattern of introgression for each locus. Estimating these cline parameters requires defining reference parental populations, estimating allele frequencies for parental populations and estimating hybrid indexes for putatively admixed individuals. The Bayesian genomic cline model requires individuals from the parental population to be specified a priori. Parental allele frequencies and hybrid indexes are estimated simultaneously with the cline parameters, and thus estimates of cline parameters reflect uncertainty in parental allele frequencies and hybrid indexes (unlike the earlier likelihood-based genomic clines). This model allows genomic clines for one or more admixed populations to be estimated with variation in cline parameters among populations. Direct estimates of population-specific clines should be useful for contrasting patterns of introgression among admixed populations or sub-populations within a hybrid zone.

Bayesian genomic cline model

For each of the parental populations, we assume that the observed allele counts ($\boldsymbol{\psi_{i0}}$ and $\boldsymbol{\psi_{i1}}$) for the ith locus follow a multinomial distribution with parameter vectors $\boldsymbol{\pi_{i0}}$ and $\boldsymbol{\pi_{i1}}$ that describe the unobserved allele frequencies for parental populations 0 and 1, respectively. We assign each of these allele frequency vectors an independent and uninformative Dirichlet prior (i.e. $\lambda_{i01} = \lambda_{i02} = \cdots = \lambda_{i0k} = 1$), which is conjugate to the multinomial likelihood. This specification results in the following analytical solution (a Dirichlet distribution) for the joint posterior probability distribution of allele frequencies for parental population 0:

$$P(\boldsymbol{\pi_0}|\boldsymbol{\psi_0}) = \prod_i \frac{1}{B(\boldsymbol{\psi_{i0}} + 1)} \prod_k \pi_{i0k}^{\psi_{i0k}}, \quad (1)$$

where the second product is taken across all k alleles for locus i and B denotes the beta function, which is the normalizing constant of a Dirichlet distribution. The posterior probability distribution of allele frequencies for parental population 1 is expressed in a similar manner.

For all potentially admixed individuals in the sample, we assume that each allele copy at each locus has ancestry from either parental population 0 or parental population 1 (i.e. the allele copy was inherited either from parental population 0 or parental population 1). Conditional on the ancestry of all allele copies in an individual ($\mathbf{z}$ below), we can model the observed allele count as a function of the allele frequencies in the appropriate parental population using multinomial distributions. This allows us to specify the following likelihood for the observed genetic data in the potentially admixed individuals ($\mathbf{x}$), which is a product across all loci, admixed populations, individuals and allele copies:

$$P(\mathbf{x}|\boldsymbol{\pi}, \mathbf{z}) = \prod_i \prod_j \prod_n \prod_a \begin{cases} \prod_k \pi_{i0k}^{x_{ijnak}} & \text{if } z_{ijna} = 0 \\ \prod_k \pi_{i1k}^{x_{ijnak}} & \text{if } z_{ijna} = 1 \end{cases} \quad (2)$$

where x_{ijnak} denotes the count of the kth allelic state in the nth individual and jth admixed population at the ith locus and ath allele copy ($\mathbf{a} = \{1,2\}$ for diploid organisms). $\mathbf{x_{ijna}}$ is a vector with a 1 for the allelic state observed and 0's for all other allelic states. $\mathbf{z}$ denotes ancestry, and z_{ijna} is 0 if individual n has ancestry from parental population 0 for the ith locus and ath allele copy and is 1 if individual n has ancestry from parental population 1 for the ith locus and ath allele copy. $\boldsymbol{\pi}$ is the parental allele frequency, as described earlier. This specification for $P(\mathbf{x}\,|\,\boldsymbol{\pi},\mathbf{z})$ parallels the admixture model proposed by Pritchard *et al.* (2000).

We propose a genomic cline function for the probability of ancestry from parental population 1 (denoted by $\boldsymbol{\phi}$; $1-\boldsymbol{\phi}$ gives the probability of ancestry in parental population 0). This function is necessary to specify the conditional probability of the ancestry vector $\mathbf{z}$, and estimating the parameters of this function is the main goal of the analysis. We begin with a polynomial function that relates hybrid index ($\mathbf{h}$) and two genomic cline parameters to an auxiliary variable $\boldsymbol{\theta}$ that is closely related to $\boldsymbol{\phi}$:

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$$\theta_{ijn} = h_n + 2(h_n - h_n^2)(\alpha_{ij} + \beta_{ij}(2h_n - 1)). \quad (3)$$

An individual with $h_n = 0$ has ancestry only from parental population 0, whereas an individual with $h_n = 1$ has ancestry only from parental population 1. The genomic cline parameter α_{ij} denotes an increase (positive values, shift of the cline to the left) or decrease (negative values, shift of the cline to the right) in the probability of parental population 1 ancestry relative to a null expectation based only on h_n, and genomic cline parameter β_{ij} denotes an increase (positive values, steeper cline) or decrease (negative values, wider cline) in the rate of transition from parental population 0 to parental population 1 ancestry along the hybrid index or genome-wide admixture gradient. Therefore, we refer to α and β as genomic cline center and rate parameters. This function is equivalent to the equation given by Szymura & Barton (1986) to relate the allele frequency for a specific locus to the mean allele frequency across loci, but with model parameter substitutions. The specific form of this function allows α to influence θ as a linear or quadratic function of h_n and β to influence θ as a linear, quadratic or cubic function of h_n. Unfortunately, when β_{ij} is negative and of a large-enough magnitude (the specific value depends on α_{ij}), Eqn 3 has the undesirable property of yielding two local extrema over the interval $0 < \theta_{ijn} < 1$. When this occurs, it is necessary to splice an additional function onto Eqn 3 to ensure that θ_{ijn} instead reaches a plateau in the vicinity of these local extrema. Formally this issue arises when $f'(h) < 0$ for some $h = \{0 \cdots 1\}$ and $f'(h = 0) > 0$ and $f'(h = 1) > 0$, where $f'(h)$ is the first derivative of Eqn 3 with respect to h. When this occurs, we splice the function $\theta_{ijn} = y$ onto Eqn 3, where y is equal to the mean value of $\boldsymbol{\theta_{ij}}$ for $A = \{h \mid f'(h) = 0\}$. $\theta_{ijn} = y$ is spliced onto Eqn 3 over the interval with lower bounds min ($M = \{h \mid f(h) = y\}$) and upper bounds max ($M = \{h \mid f(h) = y\}$), which we define as m_1 and m_2 respectively. Thus, when the afore said conditions are met, we replace Eqn 3 with:

$$\theta_{ijn} = \begin{cases} y & \text{if } m_1 \le h_n \le m_2 \\ h_n + 2(h_n - h_n^2)(\alpha_{ij} + \beta_{ij}(2h_n - 1)) & \text{otherwise} \end{cases} \quad (4)$$

We then relate θ_{ijn} to ϕ_{ijn} with the constraint that $0 \le \phi_{ijn} \le 1$:

$$\phi_{ijn} = \begin{cases} \max(0, \theta_{ijn}) & \text{if } \theta_{ijn} \le 0.5 \\ \min(1, \theta_{ijn}) & \text{if } \theta_{ijn} > 0.5 \end{cases} \quad (5)$$

$\boldsymbol{\phi}$ gives the probability of population 1 ancestry and thus specifies $P(\mathbf{z} \mid \mathbf{h}, \boldsymbol{\alpha}, \boldsymbol{\beta})$. Example genomic clines produced using Eqns 3, 4 and 5 with different values of α_{ij} and β_{ij} are shown in Fig. 1.

We define linear random effect models for both α_{ij} and β_{ij}, such that

$$\alpha_{ij} = \gamma_i + \eta_{i(j)} \quad (6)$$

and

$$\beta_{ij} = \zeta_i + \kappa_{i(j)}. \quad (7)$$

This specification assumes a locus-specific effect on each cline parameter as well as a population within locus effect. Each locus random effect is given a normal conditional prior as follows: $\gamma_i \sim N(0, \tau_\alpha)$ and $\zeta_i \sim N(0, \tau_\beta)$. Similarly, we assign normal conditional priors to each of the nested population effects, such that $\eta_{i(j)} \sim N(0, \nu_i)$ and $\kappa_{i(j)} \sim N(0, \omega_i)$, where τ_α and τ_β are precision parameters (the reciprocal of the variance) and $\boldsymbol{\nu}$ and $\boldsymbol{\omega}$ are vectors of precision parameters with one parameter for each locus (i.e. we allow the variance among admixed populations in genomic clines to vary among loci). For parameter estimation, we implemented 'sum-to-zero' constraints on the nested population effects ($\eta_{i(j)}$ and

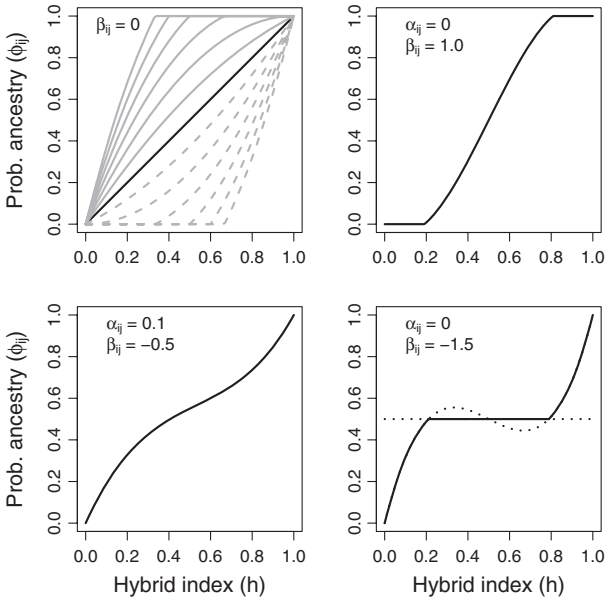


Fig. 1 Example genomic clines with different parameter values for α_{ij} (genomic cline center) and β_{ij} (genomic cline rate). When both of these values are set to 0, there is a one-to-one relationship between ϕ_{ij} and hybrid index. In the first panel, the solid black line denotes $\alpha_{ij} = 0$, solid grey lines denote α_{ij} increasing from 0.25 to 1.5, and dashed grey lines denote α_{ij} decreasing from −0.25 to −1.5. When β_{ij} is set to −1.5, it is necessary to splice functions together as described in the text. The components of the functions that were removed by splicing the functions together are shown with dotted lines.

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$\kappa_{i(j)}$) using reparameterization by sweeping as proposed by Vines *et al.* (1996). Note, when only a single admixed population is included in the analysis, $\eta_{i(j)}$, $\kappa_{i(j)}$ and the associated precision parameters are dropped from the model. To complete the Bayesian genomic cline model, we assign uninformative gamma priors to each of the precision parameters and a uniform prior to **h**. This yields the following model:

$$P(\boldsymbol{\pi_0}, \boldsymbol{\pi_1}, \mathbf{z}, \mathbf{h}, \gamma, \boldsymbol{\zeta}, \boldsymbol{\eta}, \boldsymbol{\kappa}, \tau_\alpha, \tau_\beta, \boldsymbol{\nu}, \boldsymbol{\omega} | \boldsymbol{\psi_0}, \boldsymbol{\psi_1}, \mathbf{x}) \propto$$
$$P(\boldsymbol{\pi_0}|\boldsymbol{\psi_0})P(\boldsymbol{\pi_1}|\boldsymbol{\psi_1})P(\mathbf{x}|\boldsymbol{\pi}, \mathbf{z})P(\mathbf{z}|\mathbf{h}, \boldsymbol{\alpha} = \gamma + \boldsymbol{\eta}, \boldsymbol{\beta} = \boldsymbol{\zeta} + \boldsymbol{\kappa})P(\gamma|\tau_\alpha)$$
$$P(\boldsymbol{\zeta}|\tau_\beta)P(\boldsymbol{\eta}|\boldsymbol{\nu})P(\boldsymbol{\kappa}|\boldsymbol{\omega})P(\tau_\alpha)P(\tau_\beta)P(\boldsymbol{\nu})P(\boldsymbol{\omega})P(\mathbf{h}). \quad (8)$$

It is to be noted that this joint posterior distribution is proportional to the posterior distribution for parental allele frequencies (terms one and two) and the posterior distributions for all parameters associated with the admixed populations expressed as the model likelihood, conditional priors and hyperpriors (all other terms). We obtain the joint posterior probability distribution of these parameters using Markov chain Monte Carlo (MCMC; see Appendix for details regarding the MCMC algorithm).

Simulations: genetic drift

We used the genomic cline model to quantify variation in cline parameters among genetic regions for simulated, neutrally evolving admixed populations. We modelled diploid hermaphroditic populations of fixed size and discrete nonoverlapping generations. Individuals had 10 autosomal chromosomes, each 1 Morgan in length (we assumed an average of one recombination event per Morgan per gamete). For each simulation, we seeded an admixed population with an equal number of individuals from each of two parental populations. Mating occurred randomly each generation with 80% of the gametes drawn from the admixed population and 10% of the gametes coming from each of the parental populations (i.e. $m = 0.2$). This model simulates an admixed population without spatial structure. We followed the admixed population for 25 generations. We then sampled 100 individuals from the admixed population and 50 individuals from each of the parental populations for analysis. The simulations tracked junctions along each chromosome (Fisher 1954) rather than specific allelic states, which provided complete information on ancestry across the genome. The sampled individuals from each simulation were scored for 110 bi-allelic genetic markers spaced evenly at 10-cM intervals along each chromosome (11 genetic markers per chromosome). We refer to markers by their chromosome number and map position (i.e. *c4.1* is the first marker on chromosome four). We assumed that the parental species were fixed for alternative alleles for each of these markers. We conducted 10 replicate simulations each for admixed population sizes (N) of 100, 500 and 1000. The same simulation model has been used previously to simulate admixed populations (Buerkle & Rieseberg 2008; Gompert & Buerkle 2009). Genetic data from the simulated populations are available from the Dryad repository (doi:10.5061/dryad.8623).

We estimated genomic cline parameters for each of the simulated data sets using the Bayesian genomic cline model. We based parameter estimates on 25 000 MCMC samples taken every other iteration and following a 5000 iteration burnin. Sample history plots were monitored to determine convergence of the MCMC algorithm to the stationary distribution (during the development phase of the Bayesian genomic cline model, we assessed convergence for simulated data sets under a wider set of conditions, which included variable differences in parental allele frequencies, than presented in this manuscript). Custom C++ code used for simulating data and implementing MCMC parameter estimation for the Bayesian genomic cline model, which uses the GNU Scientific Library (Galassi *et al.* 2009), is available upon request from ZG or CAB and from the Dryad repository (doi:10.5061/dryad.8623). The MCMC algorithm for each data set required approximately 4 h on a computer with two 3.0 GHz Dual-Core Intel Xeon processors and 16 GB of memory running CentosOS 5.x Linux.

Simulations: selection

We simulated additional genetic data sets to determine whether genomic cline parameters could be used to detect genetic regions affected by selection in admixed populations. Data were simulated as described in the previous section (see Simulations: genetic drift), but with a few modifications. We modelled directional selection (with incomplete dominance), underdominance, or pairwise epistasis. Directional selection can drive adaptive introgression or adaptation of an admixed population to local environmental conditions (e.g. Rieseberg *et al.* 2003; Martin *et al.* 2006; Whitney *et al.* 2006), whereas underdominance and pairwise epistasis are thought to contribute to reproductive isolation between lineages (e.g. Dobzhansky 1937; Gavrilets *et al.* 1998; Raufaste *et al.* 2005; Nolte *et al.* 2009). We assumed additive fitness effects for all models, where relative fitness was $f = 1 - x_1 - x_2$. For directional selection and underdominance, x_1 and x_2 were a function of an individual's genotype at two focal loci (*c1.6* and *c2.6*)

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located in the center of chromosomes 1 and 2. Specifically, with directional selection x_1 was 0 (homozygotes for population 0 ancestry), $\frac{s}{2}$ (heterozygotes), or s (homozygotes for population 1 ancestry) based on the individual's genotype at locus *c1.6* and x_2 was calculated similarly, but with respect to locus *c2.6*. For underdominance, we defined fitness similarly but with x_1 and x_2 set to 0 for homozygote genotypes and s for heterozygotes. We modelled pairwise epistasis with fitness determined by one (*c1.6*, *c2.6*) or two pairs (*c1.4*, *c1.8* and *c2.4*, *c2.8*) of interacting loci. Specifically, x_1 was s for multilocus genotypes homozygous for population 0 ancestry at one locus and homozygous for population 1 ancestry at the other interacting locus, $\frac{s}{2}$ for multilocus genotypes heterozygous at one locus and homozygous for ancestry from either parental population at the interacting locus, and 0 for all other multilocus genotypes. When we modelled two pairs of interacting loci, we defined x_2 similarly to x_1, but based on the second pair of loci. For each form of selection, we simulated data with s set to 0.1, 0.2, or 0.4. We modelled viability selection (i.e. reduced probability of survivorship of zygotes). We assumed an admixed population size of 500 individuals that had experienced admixture for 25 generations. We simulated data with dispersal rates of 0.2 (as used for simulations without selection) and 0.05 (i.e. 2.5% of gametes sampled from each parental population) to determine whether dispersal rate from the parental populations affected our ability to discriminate between selected and neutral genetic regions. We simulated 10 replicate data sets for each model (i.e. form and intensity of selection).

We estimated genomic cline parameters for these simulated data sets as described in the previous section (see Simulations: genetic drift). An aim of these simulations was to determine whether cline parameters estimated for loci affected by selection were extreme relative to neutral loci. To address this question, we designated statistical 'outlier' loci based on estimates of locus-specific cline parameters relative to their random effect conditional prior, which can be thought of as a genome-wide distribution for the cline parameter (Gompert *et al.* 2011). We designated 'outlier' loci with respect to each cline parameter: α (genomic cline center) and β (genomic cline rate). For each simulated data set, we defined locus i as a statistical 'outlier' with respect to α if the posterior estimate (i.e. the median of the posterior distribution) of α_i was not contained in the interval q_N, where q_N is the interval bounded by the $\frac{N}{2}$ and $\frac{1-N}{2}$ quantiles of the posterior probability distribution for the random-effect prior for $\boldsymbol{\alpha}$ (based on the posterior probability distribution for τ_α). We defined 'outlier' loci with respect to β similarly. We specified $N = 0.05$ for 'outlier' designation in this manuscript. Defining a boundary for the designation of 'outlier' loci is subjective, and we do not directly equate 'outliers' with loci experiencing selection (see Discussion). We calculated the number of statistical 'outliers' that corresponded to the focal loci experiencing selection, loci within 10 cM of the focal loci (*c1.5*, *c1.7*, *c2.5* and *c2.7*, or *c1.3*, *c1.5*, *c1.7*, *c1.9*, *c2.3*, *c2.5*, *c2.7* and *c2.9*), and other loci. We then calculated a false discovery rate for each set of model conditions as the ratio of the number of statistical 'outliers' >10 cM from the selected loci to the total number of 'outliers'.

Simulations: multiple admixed populations

We simulated a single data set to determine whether including the nested population effects in the Bayesian genomic cline model could be used to identify genetic regions experiencing geographically variable selection. We modified the data simulation procedure in a few ways. We simulated six admixed populations each with 500 individuals. There was no gene flow among admixed populations, but we set $m = 0.2$ between parental populations and each admixed populations. For three admixed populations, we simulated directional selection on *c1.6* and *c2.6*, with $s = 0.4$. The other three populations did not experience selection. MCMC was used to estimate genomic cline parameters for the combined data set (all six admixed populations), such that we obtained estimates of α_{ij} and β_{ij} for each of the 110 loci (i) and each of the six populations (j). We also examined posterior estimates of the nested population effects $\eta_{i(j)}$ and $\kappa_{i(j)}$ to determine whether these effects were larger for the loci experiencing selection.

Mus *hybrid zones*

We used the Bayesian genomic cline model to estimate cline parameters for SNP genetic markers from two transects through a central European hybrid zone between *M. domesticus* and *M. musculus* (data from Teeter *et al.* 2008 and Teeter *et al.* 2010). The southern transect, which we refer to as the Bavarian admixed population, is located in the German state of Bavaria and western Austria and has been studied extensively (Payseur *et al.* 2004; Teeter *et al.* 2008, 2010). The northern transect, which we refer to as the Saxony admixed population, is located in the German states of Thuringia, Saxony-Anhalt, and Saxony and was first reported by Teeter *et al.* (2010). These data include 38 autosomal SNPs (two to three SNPs on each of the 19 autosomal chromosome pairs) and three X-linked SNPs genotyped for 449 mice from the Bavarian admixed population and 321 mice from the Saxony admixed population. Ten parental *M. domesticus* and *M. musculus* were initially

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genotyped for these SNP markers and were found to be fixed for different alleles (Payseur *et al.* 2004; Teeter *et al.* 2008). We used these data to estimate parental allele frequencies. We analysed the Bavarian and Saxony data in a single model, which included a nested population effect for each cline parameter (see Eqns. 6 and 7). We treated each transect rather than each sampling locality as a distinct admixed population because many sampling localities had very low sample sizes (one or two mice; Teeter *et al.* 2008, 2010) and because our primary interest was comparing patterns of introgression between the transects. We estimated cline parameters using MCMC based on a single chain yielding 25 000 samples from the posterior probability distribution (taken every other iteration), which we obtained following a 5000 iteration burnin. We assessed convergence to the stationary distribution by monitoring sample history plots, and more formally using Gelman and Rubin's convergence diagnostic based on the model deviance (Gelman & Rubin 1992). The latter was calculated in R using the *coda* package (Plummer *et al.* 2006) and involved MCMC output from a second chain. We designated 'outlier' loci as described in a previous section (Simulations: selection), but with respect to the locus effects for cline center (γ_i) and cline rate (ζ_i) rather than α and β as these parameters cannot be equated with each other when population random effects are included.

Results

Simulations

We first simulated data and estimated parameters using the Bayesian genomic cline model in the absence of selection. Inspection of sample history plots for MCMC parameters indicated convergence to the stationary distribution. We obtained precise and very accurate estimates of hybrid index for the admixed individuals (i.e. correlations between true hybrid indexes and point estimates from the genomic cline model were generally ≥ 0.99). Estimated genomic clines varied among loci. This variation was more pronounced with $N = 100$ than with population sizes of 500 or 1000 (Fig. 2). Estimates of α_i varied to a greater extent among loci than estimates of β_i, indicating that genetic drift shifted the mean expected ancestry at individual loci more than the rate of transition in the probability of ancestry. For example, with $N = 100$ minimum and maximum estimates of α_i for data from the first replicate simulation were −0.267 (95% equal tail probability interval [ETPI] −0.403 to −0.130) and 0.272 (95% ETPI 0.133–0.409), whereas minimum and maximum estimates of β_i were −0.157 (95% ETPI −0.483 to 0.059) and 0.165 (95% ETPI −0.068 to 0.504). Consistent with these results, we obtained higher estimates of among-locus variation in cline parameters (i.e. the reciprocal of τ_α and τ_β) with decreased population sizes, and had greater uncertainty and lower estimates for $\frac{1}{\tau_\beta}$ than $\frac{1}{\tau_\alpha}$ (Fig. 3).

The second set of simulations included selection. Both directional selection and underdominance affected genomic cline parameters. For these forms of selection, the number of genetic regions classified as 'outliers' (i.e. genetic regions with extreme estimates of α_i or β_i) was greater when stronger selection was simulated (Table 1). For example, when data were simulated with $s = 0.4$ and $m = 0.05$, all loci for the ten replicate simulations that were direct targets of selection were identified as 'outliers', whereas 11 (55%; directional selection) or three (15%; underdominance) of these loci were classified as 'outliers' for data simulated with $s = 0.1$. The effect of dispersal rate on genomic cline parameters was

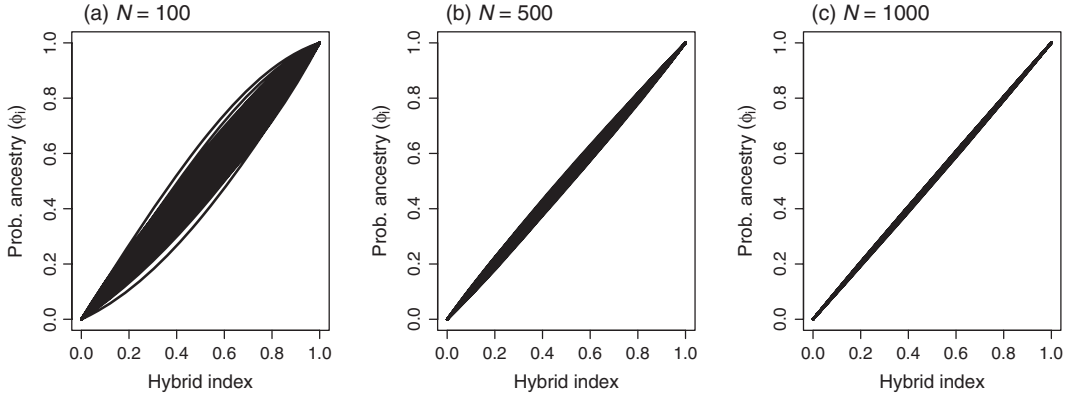


Fig. 2 Genomic clines from three simulated admixed populations in the absence of selection (population size (N) of 100, 500 and 1000). Point estimates of genomic clines, which are based on the median from the posterior distribution, are shown for each of 110 loci.

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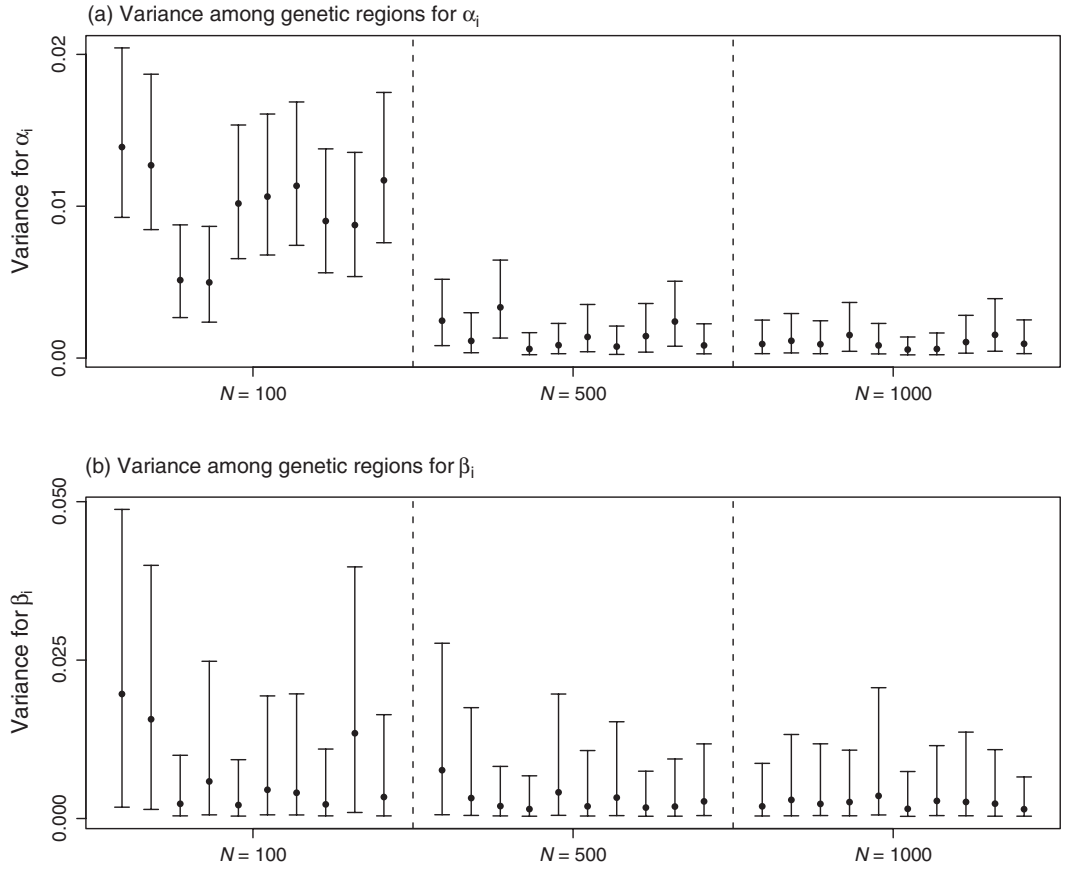


Fig. 3 Variance among genetic regions in α_i (a) and β_i (b) cline parameters in simulations of neutral admixture. Points and error bars give median and 95% equal tail probability intervals (ETPI) for $\frac{1}{\tau_\alpha}$ and $\frac{1}{\tau_\beta}$ for each replicate simulation. Dashed lines separate groups of 10 replicates with admixed population sizes of 100, 500 and 1000.

more complex (Figs 4 and 5). For directional selection, a greater number of loci were classified as 'outliers' with a low dispersal rate ($m = 0.05$) than a high dispersal rate ($m = 0.2$), but this effect was observed for both the loci that were direct targets of selection and other non-target loci resulting in a higher false discovery rate (Table 1). Regardless of the dispersal rate, loci affected by directional selection were generally classified as 'outliers' with respect to the cline parameter α, as would be expected. With underdominance, the dispersal rate determined whether loci affected by selection were likely to be classified as 'outliers' with respect to α or β. When the dispersal rate was low, selected loci tended to be classified as 'outliers' with respect to α, whereas with a high dispersal rate these loci were classified as 'outliers' with respect to β. The effect of underdominance when the dispersal rate was low can be explained by an initial shift in the frequency of ancestry for the selected genetic region in the admixed population because of drift, which causes a greater proportion of alleles inherited from one parental population to occur in inter-population heterozygote genotypes than alleles inherited from the other parental population. Alleles that occur less frequently in inter-population heterozygote genotypes will have a higher marginal fitness and homozygotes of this type will increase relative to other genotypes (as in directional selection). Conversely, pairwise epistasis did not have a major effect on genomic cline parameters (Table 1, Figs 4 and 5). A relatively small fraction of genetic regions affected by epistatic selection (no more than 10% with respect to α and no more than 30% with respect to β) was classified as 'outliers'. Moreover, with epistasis the proportion of selected genetic regions classified as 'outliers' was not strongly dependent on the strength of selection, the number genetic regions affecting fitness, or the dispersal rate from parental populations.

With false positives defined as loci more than 10 cM from the actual targets of selection, when we modelled directional selection or underdominance the false discovery rate specifying the fraction of 'outlier' loci unassociated with variation in fitness was low (i.e. < 0.1; Table 1). However, with weak selection and a low dispersal rate, the false discovery rate was high (i.e. 0.38–

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Table 1 Counts of 'outlier' loci from simulated data

Selection	m	s	$n_0(\alpha)$	$n_1(\alpha)$	$n_2(\alpha)$	$n_0(\beta)$	$n_1(\beta)$	$n_2(\beta)$	FDR
Directional	0.05	0.1	11 (0.55)	4 (0.10)	9 (0.01)	0 (0.00)	0 (0.00)	0 (0.00)	0.38
		0.2	20 (1.00)	6 (0.15)	4 (0.00)	0 (0.00)	0 (0.00)	0 (0.00)	0.13
		0.4	20 (1.00)	8 (0.20)	3 (0.00)	2 (0.10)	0 (0.00)	1 (0.00)	0.09
	0.2	0.1	0 (0.00)	0 (0.00)	0 (0.00)	0 (0.00)	0 (0.00)	0 (0.00)	0.00
		0.2	2 (0.10)	0 (0.00)	0 (0.00)	0 (0.00)	0 (0.00)	0 (0.00)	0.00
		0.4	6 (0.30)	5 (0.13)	1 (0.00)	0 (0.00)	0 (0.00)	0 (0.00)	0.08
Underdom.	0.05	0.1	3 (0.15)	1 (0.03)	11 (0.01)	0 (0.00)	0 (0.00)	0 (0.00)	0.73
		0.2	8 (0.40)	1 (0.03)	19 (0.02)	0 (0.00)	0 (0.00)	0 (0.00)	0.68
		0.4	20 (1.00)	12 (0.30)	1 (0.00)	0 (0.00)	0 (0.00)	0 (0.00)	0.03
	0.2	0.1	0 (0.00)	0 (0.00)	0 (0.00)	0 (0.00)	0 (0.00)	0 (0.00)	0.00
		0.2	0 (0.00)	0 (0.00)	0 (0.00)	0 (0.00)	0 (0.00)	0 (0.00)	0.00
		0.4	1 (0.05)	0 (0.00)	0 (0.00)	19 (0.95)	6 (0.15)	1 (0.00)	0.04
Epi. (1p)	0.05	0.1	2 (0.10)	2 (0.05)	82 (0.08)	0 (0.00)	2 (0.05)	44 (0.04)	0.95
		0.2	1 (0.05)	1 (0.03)	35 (0.03)	2 (0.10)	2 (0.05)	50 (0.04)	0.93
		0.4	2 (0.10)	2 (0.05)	34 (0.03)	6 (0.30)	4 (0.10)	70 (0.07)	0.88
	0.2	0.1	1 (0.05)	0 (0.00)	47 (0.04)	4 (0.20)	2 (0.05)	42 (0.04)	0.93
		0.2	0 (0.00)	2 (0.05)	52 (0.05)	2 (0.10)	2 (0.05)	50 (0.05)	0.94
		0.4	2 (0.10)	3 (0.08)	31 (0.03)	0 (0.00)	2 (0.05)	56 (0.05)	0.93
Epi. (2p)	0.05	0.1	1 (0.03)	3 (0.04)	34 (0.03)	4 (0.10)	4 (0.05)	28 (0.03)	0.84
		0.2	3 (0.08)	2 (0.03)	57 (0.06)	0 (0.00)	0 (0.00)	34 (0.03)	0.95
		0.4	1 (0.03)	0 (0.00)	18 (0.02)	4 (0.10)	6 (0.08)	48 (0.05)	0.86
	0.2	0.1	0 (0.00)	4 (0.05)	45 (0.05)	4 (0.10)	0 (0.00)	42 (0.04)	0.92
		0.2	2 (0.05)	4 (0.05)	35 (0.04)	2 (0.05)	8 (0.10)	34 (0.03)	0.81
		0.4	2 (0.05)	3 (0.04)	58 (0.06)	4 (0.10)	2 (0.03)	36 (0.04)	0.90

m and s give the dispersal rate and selection coefficient for the simulated data as described in the main text. For both genomic cline parameters (α and β), we report the counts of 'outlier' loci (summed across 10 replicates) that correspond to loci directly affected by selection (n_0), loci within 10 cM of the affected loci (n_1), and other loci (n_2). Following each count, in parentheses we report the proportion of loci of each type classified as 'outliers'. Counts and proportions for n_2 correspond to false positives. FDR denotes the false discovery rate (i.e. the fraction of loci recognized as 'outliers' that were not directly subject to selection or with 10 cM of the selected loci).

0.73 of 'outlier' loci were unassociated with targets of selection). Regardless of the false discovery rate, 'outliers' were enriched for loci under selection (truly selected and neighbouring loci). Specifically, the mean proportion of target loci classified as 'outliers' was 0.45, whereas the proportion of off-target loci (i.e. loci >10 cM from selected target loci) never exceeded 0.02. Thus, even when the false discovery rate was relatively high, the probability of classifying off-target loci as 'outliers' (false positives; typically referred to as a type I error) was still very low. A similar, albeit weaker signal was observed for loci within 10 cM of the selected locus (Table 1). This result is consistent with estimates of cline parameters along the simulated chromosomes, with extreme cline parameter estimates often centered on loci directly affected by selection and decreasing in magnitude as a function of distance from the target loci (Fig. 5).

Results with pairwise epistasis were noticeably different. Although the proportion of off-target loci classified as 'outliers' was generally quite low and never exceeded 0.08, the false discovery rate was always very high (between 0.81 and 0.95; Table 1). Unlike our findings for directional selection and underdominance, with pairwise epistasis target and near-target loci were at best modestly enriched for loci under selection. For example, the mean proportion of target loci classified as 'outliers' was 0.07, whereas the mean proportion of off target loci classified as 'outliers' was 0.04.

We also simulated a single data set with six admixed populations, three of which experienced selection at two focal loci. Our analysis of this data set indicated that the Bayesian genomic cline model can be used to detect geographically variable selection. Specifically, for both selected loci, although the signal was more clear for *c2.6*, we saw large negative estimates of the locus effect γ_i (−0.10 and −0.12) coupled with large negative estimates of the nested effect $\eta_{i(j)}$ for populations that experienced selection (*c1.6*: −0.045, −0.085 and −0.081; *c2.6*: −0.182, −0.107 and −0.134) and moderate to large positive estimates of $\eta_{i(j)}$ for populations that did not experience selection (*c1.6*: 0.027, 0.037 and 0.149; *c2.6*: 0.074, 0.147 and 0.202; Fig. 6). This resulted in extreme negative estimates of α_{ij} for populations that experienced selection, but mod-

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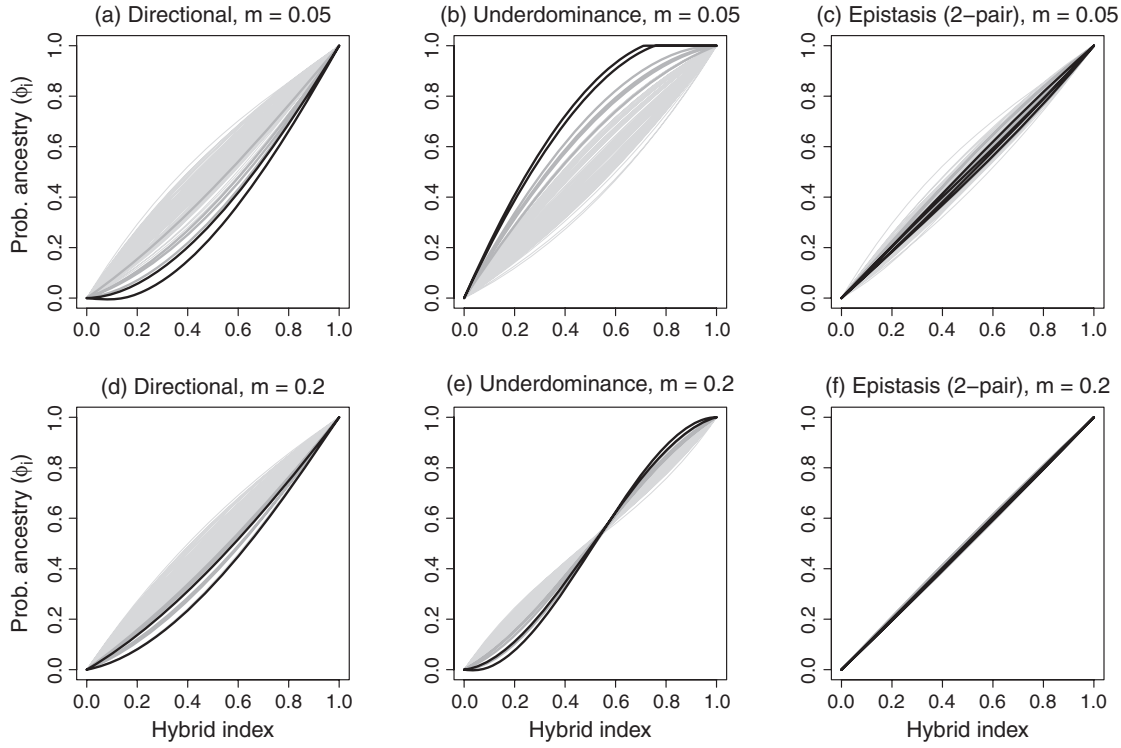


Fig. 4 Genomic clines from simulated admixed populations in which two loci experienced selection. Point estimates of genomic clines, which are based on the median from the posterior distribution, are shown for each of 110 loci. Genomic clines for the two or four selected loci are in black, genomic clines for loci within 10 cM of the selected loci are in dark grey, and the remaining clines are in light grey. Each plot gives results from a single replicate with $s = 0.4$, directional selection, underdominance, or epistasis with two pairs of loci, and $m = 0.05$ or 0.2.

erately positive to near zero estimates of α_{ij} for populations that did not experience selection. We observed comparably large population effects for several other loci (e.g. locus *c4.1*; Fig. 6), but these deviations did not generally involve all populations and were not associated with extreme estimates of the locus effect γ_i.

Mus *hybrid zones*

Similar to MCMC results from simulated data, sample history plots and Gelman and Rubin's convergence diagnostic indicated convergence of the MCMC to the stationary distribution (potential scale reduction factor $= 1.00$, 97.5% quantile $= 1.01$). Genomic clines for the *Mus* hybrid zones varied among loci (Table 2, Fig. 7 and Fig. S1, Supporting information). For example, genomic clines for several loci were very steep and suggested limited introgression between *M. domesticus* and *M. musculus* (e.g. Bavaria X.089Pola1 $\beta_{ij} = 3.142$ 95% ETPI [2.101–4.866]), whereas genomic clines for other loci indicated considerably greater rates of introgression (e.g. Saxony 16.014 $\beta_{ij} = -1.805$ 95% ETPI [−2.145 to −1.458]). Genomic clines for some loci, such as 3.007 in the Bavarian admixed population, indicated an excess of *M. musculus* ancestry in admixed individuals ($\alpha_{ij} =$ 0.933 95% ETPI [0.650–1.165]). A single locus (17.09) was designated as an 'outlier' with respect to γ_i (i.e. the locus effect for α_{ij}). Estimates of α_{ij} for this locus in both admixed populations were negative and of large magnitude, indicating an excess of *M. domesticus* ancestry. Two loci (11.089, X.089Pola1) were designated 'outliers' with respect to ζ_i (i.e. the locus effect for β_{ij}). Estimates of β_{ij} for these loci were high and positive in both admixed populations, consistent with steep genomic clines and low rates on introgression between *M. domesticus* and *M. musculus* (Table 2).

We detected both discordance and concordance in cline parameters between the Bavarian and Saxony admixed *Mus* populations. For example, genomic clines were nearly identical for some loci (e.g. 1.046), whereas other loci had more dissimilar clines owing to differences in α_{ij} (e.g. 13.056), β_{ij} (e.g. 5.007, 16.014), or both (e.g. 2.078; Table 2, Fig. 7 and Fig. S1, Supporting information). Finally, we found similarities and differences between the estimated genomic clines and previous estimates of geographic cline width and centre (Teeter *et al.* 2008, 2010). Estimates of geographic cline widths for two X-linked markers (X.070Emd and

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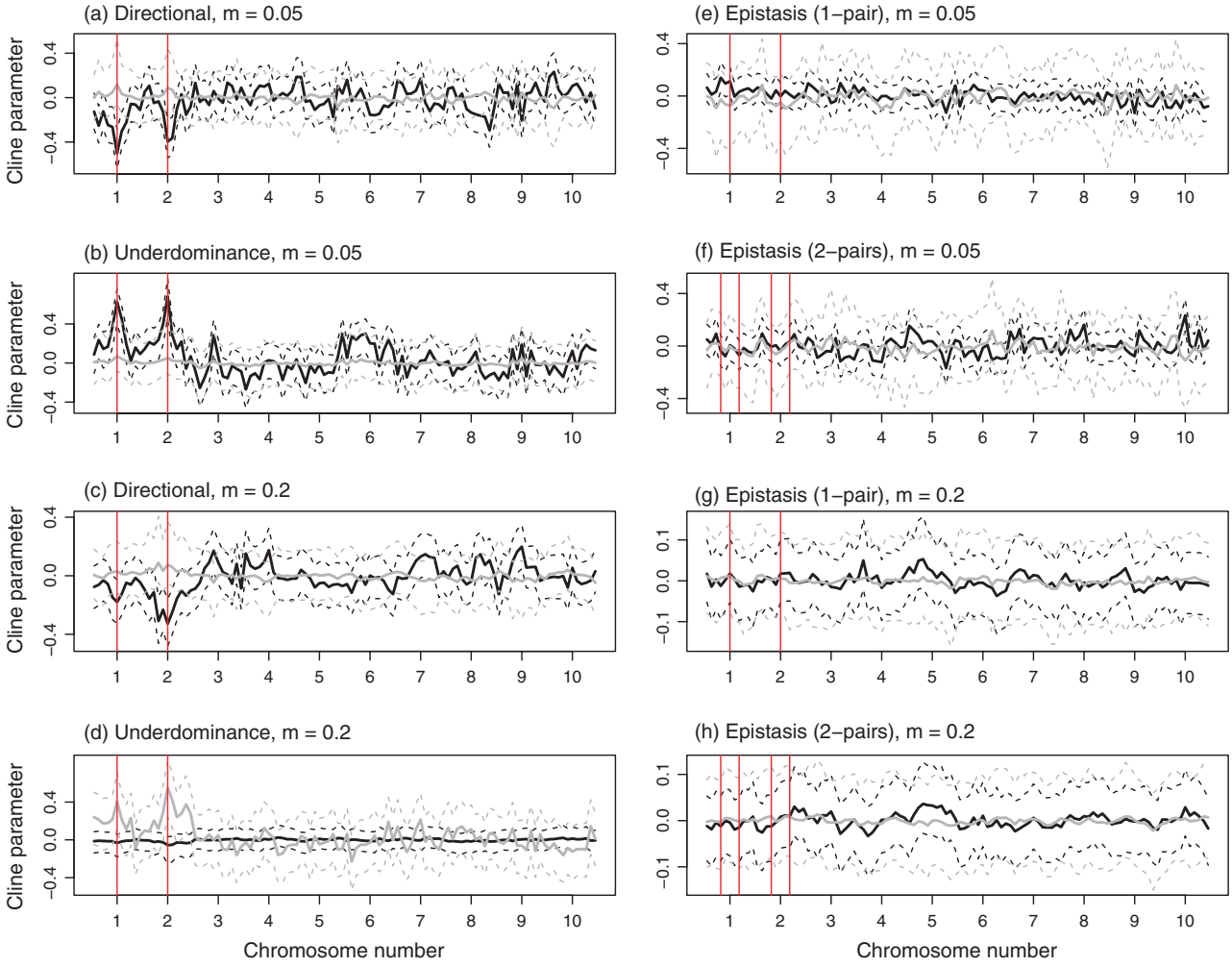


Fig. 5 Genomic cline parameter estimates as a function of genetic map location for simulated admixed populations. Solid lines denote the point estimate for α_i (black) and β_i (grey) for each locus, and dashed lines give the 95% ETPI for these estimates. Genome location is given on the *x*-axis with integers designating the center of each chromosome. Vertical red lines designate the targets of selection. Each plot depicts a single replicate for $s = 0.4$).

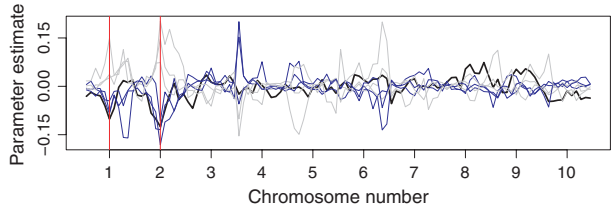


Fig. 6 Genomic cline parameter estimates for six simulated admixed populations as a function of genetic map location. Lines denote the point estimate for the locus-specific effect γ_i (black) and the nested population effects $\eta_{i(j)}$ for the three populations experiencing selection (blue) and the three populations not experiencing selection (grey). Genome location is given on the *x*-axis with integers designating the center of each chromosome. Vertical red lines designate the targets of selection.

X.089Pola1) were narrow in both admixed populations, which was matched by steep genomic clines and large positive estimates of β_{ij} in our analysis. Several autosomal loci also yielded geographic and genomic cline estimates that were consistent with one another, such as 9.075 and 16.014 in the Bavarian admixed population. Conversely, a narrow geographic cline width for locus 4.057 in both admixed populations was associated with near zero estimates of β_{ij}, suggesting that estimates of geographic and genomic cline can differ from one another.

Discussion

We demonstrated the efficacy of a new hierarchical Bayesian model for estimating genomic clines from simulated and empirical data. Consistent with our expectations, we detected increased variation among genetic regions for estimated genomic cline parameters in smaller admixed populations. Directional selection and

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Table 2 Estimates of geographic and genomic cline parameters for the mouse hybrid zone

	Bavaria				Saxony			
Locus	Centre (km)	Width (km)	α_{ij} (CI)	β_{ij} (CI)	Centre (km)	Width (km)	α_{ij} (CI)	β_{ij} (CI)
1.014	60.9	18.5	−0.27 (−0.45, −0.14)	1.06 (0.79, 1.46)	12.4	60	−0.71 (−1.04, −0.47)	0.98 (0.58, 1.35)
1.046	75.7	81.9	−0.39 (−0.52, −0.27)	1.05 (0.82, 1.30)	11.8	44.3	−0.42 (−0.60, −0.27)	1.03 (0.77, 1.32)
1.159	52.1	36.1	0.41 (0.3, 0.52)	0.39 (0.21, 0.59)	12	79.4	−0.36 (−0.50, −0.21)	0.46 (0.24, 0.71)
2.030	58.4	20.7	−0.03 (−0.17, 0.08)	1.09 (0.86, 1.36)	10.5	13.8	−0.22 (−0.67, 0.07)	2.77 (1.83, 4.13)
2.078	50.5	74.9	0.39 (0.26, 0.54)	0.56 (0.29, 1.00)	10.7	29.2	−0.33 (−1.04, −0.13)	1.28 (0.80, 2.47)
2.165	61.7	21.1	−0.31 (−0.46, −0.19)	1.20 (0.95, 1.49)	11.1	38.3	−0.36 (−0.54, −0.22)	1.18 (0.93, 1.48)
3.007	30.9	10.9	0.93 (0.65, 1.16)	0.34 (−0.49, 1.01)	10.6	44.3	−0.23 (−0.43, −0.04)	0.95 (0.54, 1.39)
3.140	69.9	65.1	−0.32 (−0.47, −0.19)	1.08 (0.84, 1.36)	11.2	41.1	−0.33 (−0.48, −0.19)	1.06 (0.80, 1.35)
4.057	99.9	14.4	−0.56 (−0.77, −0.43)	0.18 (−0.08, 0.84)	16.7	12.4	−0.73 (−1.26, −0.58)	−0.04 (−0.38, 0.31)
4.129	57.5	13	−0.09 (−0.28, 0.14)	2.18 (1.51, 3.31)	10.1	17.7	−0.18 (−0.48, 0.06)	2.30 (1.67, 3.42)
5.007	53.9	45.8	0.31 (0.19, 0.43)	0.28 (0.06, 0.49)	96.1	29.6	0.29 (0.09, 0.57)	1.32 (0.87, 2.10)
5.097	57.7	6.5	−0.77 (−1.04, −0.5)	3.84 (2.95, 4.91)	13.3	61.4	−0.82 (−1.22, −0.66)	0.18 (−0.24, 0.90)
6.088	57.9	15.4	−0.13 (−0.28, 0)	1.45 (1.15, 1.83)	10.4	36.1	−0.21 (−0.38, −0.07)	1.02 (0.73, 1.36)
6.113	62	26.3	−0.26 (−0.4, −0.14)	1.15 (0.92, 1.46)	10.4	44	−0.22 (−0.36, −0.09)	0.95 (0.65, 1.23)
7.083	11.2	17.4	−0.63 (−0.74, −0.53)	−0.07 (−0.29, 0.15)	11.6	80.7	−0.08 (−0.23, 0.05)	0.62 (0.38, 0.85)
7.126	68.7	62	−0.23 (−0.35, −0.11)	0.85 (0.61, 1.09)	11.5	64.3	−0.18 (−0.33, −0.03)	0.42 (0.13, 0.74)
8.078	95.1	17.4	−0.41 (−0.53, −0.3)	0.62 (0.40, 0.85)	10.4	35	−0.22 (−0.39, −0.05)	0.89 (0.60, 1.35)
8.101	58.1	12.9	−0.15 (−0.33, −0.02)	1.29 (1.00, 1.65)	97.8	23.4	0.01 (−0.15, 0.28)	1.41 (1.11, 2.20)
9.052	56.3	14.8	0.06 (−0.11, 0.24)	1.28 (0.92, 2.00)	10.5	28.7	−0.02 (−0.38, 0.22)	2.84 (1.44, 4.25)
9.075	56.3	6.4	−0.18 (−0.48, 0.04)	2.80 (2.12, 4.08)	73.3	66.5	0.83 (0.71, 1.00)	−0.23 (−0.55, 0.13)
10.045	63.6	13.9	0.06 (−0.08, 0.18)	0.12 (−0.11, 0.37)	12	68.9	−0.62 (−1.11, −0.41)	1.00 (0.55, 1.88)
10.055	–	–	−0.73 (−0.95, −0.52)	2.02 (1.55, 2.50)	–	–	−0.52 (−0.88, −0.02)	2.45 (1.78, 4.84)
11.053	60.7	28.6	−0.17 (−0.35, −0.03)	1.19 (0.92, 1.57)	12.8	87.7	−0.51 (−0.94, −0.20)	0.29 (−0.25, 1.41)
11.089	58.2	10.1	−0.93 (−1.45, −0.15)	4.14 (1.38, 6.31)	10.8	32.1	−0.95 (−1.78, −0.19)	2.38 (1.08, 4.13)
12.031	11.5	12.2	−1.28 (−1.62, −1.04)	1.53 (1.05, 2.07)	11.7	49.3	−0.19 (−0.45, 0.07)	1.59 (1.05, 2.25)
12.099	57.1	23.6	0.06 (−0.14, 0.22)	0.97 (0.67, 1.37)	11.1	45.1	−0.31 (−0.47, −0.15)	0.96 (0.69, 1.29)
13.029	17.4	34.8	−0.75 (−0.84, −0.66)	−0.08 (−0.28, 0.11)	15	72.2	−1.09 (−1.51, −0.80)	−0.10 (−0.42, 0.18)
13.056	54.6	26.5	0.27 (0.15, 0.39)	0.52 (0.33, 0.71)	10.3	52.5	−0.15 (−0.30, −0.00)	0.57 (0.36, 0.79)
14.031	69.9	95.6	−0.12 (−0.24, 0)	0.52 (0.31, 0.73)	11.2	55.3	−0.63 (−1.05, −0.39)	1.19 (0.64, 1.88)
14.074	78.5	84.4	−0.54 (−0.73, −0.38)	1.32 (1.02, 1.71)	11.4	37.4	−0.57 (−0.78, −0.41)	1.35 (1.05, 1.74)
15.065	89.1	91.2	−0.91 (−1.16, −0.67)	1.48 (0.91, 1.99)	13.1	76.3	−0.52 (−0.79, −0.38)	0.81 (0.46, 1.30)
15.099	61	25.7	−0.12 (−0.26, 0)	0.93 (0.70, 1.26)	10.9	20.7	−0.12 (−0.32, 0.20)	2.07 (1.18, 3.59)
16.014	56.5	7	−0.09 (−0.25, 0.05)	2.06 (1.61, 2.65)	10.2	18.4	−0.05 (−0.17, 0.06)	−1.80 (−2.14, −1.45)
17.046	54.6	14.7	0.27 (0.16, 0.39)	0.72 (0.53, 0.93)	13.9	93.1	−0.39 (−0.56, −0.24)	0.72 (0.50, 0.95)
17.091	87.2	54.9	−1.54 (−1.98, −1.22)	2.37 (1.65, 3.11)	13.2	74.7	−1.50 (−1.98, −1.13)	2.36 (1.63, 3.10)
18.028	10	13.4	−0.73 (−1, −0.56)	0.64 (0.24, 1.22)	13.7	84.5	−0.65 (−0.94, −0.50)	0.73 (0.37, 1.25)
18.064	61.7	20.6	−0.4 (−0.6, −0.24)	1.59 (1.26, 2.11)	11	37	−0.44 (−0.65, −0.27)	1.57 (1.23, 2.00)
19.044	58.4	16.2	−0.2 (−0.33, −0.07)	1.08 (0.86, 1.34)	10.4	25.6	−0.10 (−0.25, 0.05)	1.12 (0.91, 1.43)
19.052	75	72.2	−0.38 (−0.57, −0.24)	0.78 (0.53, 1.13)	13.5	86.9	−0.70 (−0.99, −0.49)	0.79 (0.50, 1.16)
X.070Emd	55.4	5.1	−0.19 (−0.46, 0.1)	2.67 (1.97, 6.67)	10.4	18.4	−0.28 (−0.69, 0.05)	2.70 (1.99, 6.72)
X.089PolaI	55	4.4	−0.05 (−0.36, 0.24)	3.14 (2.10, 4.86)	–	–	−0.17 (−0.64, 0.16)	3.17 (2.09, 4.93)
X.099Xist	10.8	29.2	−0.5 (−0.65, −0.36)	0.21 (−0.06, 0.51)	10.5	15.2	−0.49 (−0.80, −0.15)	3.38 (1.92, 5.73)

Geographic cline estimates are taken from Teeter *et al*. (2008, 2010). Correlations between geographic and genomic cline parameters were: Bavaria centre × $\alpha_{ij} = -0.034$ ($P = 0.837$), Bavaria width × $\beta_{ij} = -0.240$ ($P = 0.136$), Saxony centre × $\alpha_{ij} = 0.483$ ($P = 0.002$), Saxony width × $\beta_{ij} = -0.395$ ($P = 0.012$).

underdominance had detectable and predictable effects on genomic cline parameters, with the magnitude of the effect proportional to the strength of selection. When simulated populations experienced directional selection or underdominance, the false discovery rate for classifying genetic regions distant from targets of selection as 'outlier' loci was generally low (i.e. <0.1). This is an improvement relative to the likelihood-based genomic cline method developed by Gompert & Buerkle (2009). Nonetheless, with weak selection and a low dispersal rate, hybrid zone dynamics were dominated by genetic drift causing a considerably higher false discovery rate.

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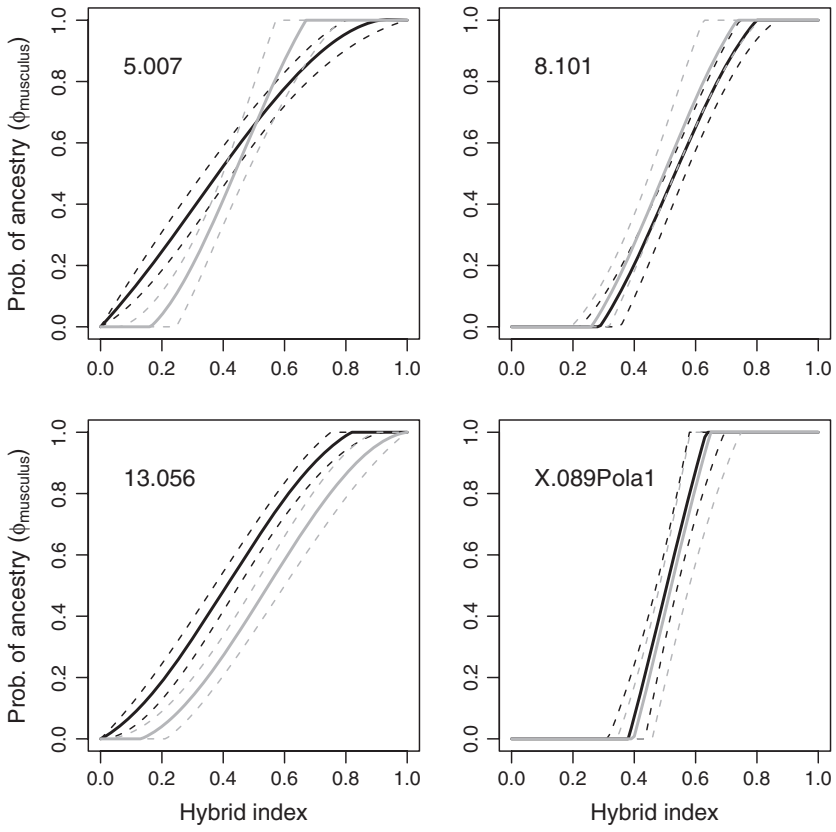


Fig. 7 *Mus* genomic clines for four typical loci (genomic clines for all loci are depicted in Fig. S1, Supporting information). Solid lines denote point estimates of genomic clines for the Bavarian (black) and Saxony (grey) admixed populations. 95% ETPI for these cline estimates are denoted with dashed lines. The name of each locus (chromosome number followed by map location) is also given.

However, even under these circumstances, 'outlier' loci were clearly enriched for genetic regions targeted by selection and false positives (type I errors) were rare. However, the effect of pairwise epistasis on genome-wide patterns of introgression and genomic cline parameters, whether involving one or two pairs of interacting loci, was minimal. The false discovery rate for 'outliers' when fitness was determined by pairwise epistasis was very high, although the false positive rate remained quite low and 'outlier' loci were slightly enriched for genetic regions targeted by selection. We were able to detect a signal of geographically variable selection (underdominance) in simulated data. Specifically, for loci experiencing selection in some but not all geographic regions, we obtained large estimates of nested population effects coupled with large locus effects for cline parameters.

The Bayesian genomic cline model we have developed will facilitate the study of speciation by providing a general framework for mapping the genetic architecture of reproductive isolation between divergent hybridizing lineages. For example, if speciation occurs because of genomically widespread underdominance driven by multifarious divergent selection affecting many traits, many genetic regions should display moderate associations with reproductive isolation (i.e. many genetic regions with moderate estimates of α_i or β_i; Nosil & Sandoval 2008; Nosil *et al.* 2009). Alternatively, if speciation occurs because of strong selection and differentiation related to a small number of traits and genes, fewer genetic regions should be associated with reproductive isolation, but these few regions should have more extreme genomic clines (i.e. large estimates of α_i, or large positive estimates of β_i). Regardless of the number of loci that contribute to isolation, it is important to note that the effect of selection on genomic cline parameters can be rather complex. Specifically, underdominance can affect either α or β depending on the rate of dispersal from parental populations. This means that 'outliers' with respect to α could be associated with directional selection or underdominance, and additional analyses or experiments would be required to determine the specific cause.

Unfortunately, our results indicate that the Bayesian genomic cline model might be less useful for detecting pairwise epistasis, which is thought to be a major component of reproductive isolation (Gavrilets *et al.* 1998; Coyne & Orr 2004). This is because, at least for the simulation conditions we explored, pairwise epistasis truly had little effect on the rate or pattern of introgression for the target loci, and thus did not affect genomic cline parameters. We have obtained similar results previously when modelling genome-wide epistasis (Gompert & Buerkle 2009). These findings suggest the epistatic inter-

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actions that do not affect the fitness of double heterozygotes might not always serve as effective barriers to gene flow, although additional theoretical work is necessary to thoroughly explore this possibility and the underlying dynamics. Whereas pairwise epistasis did not affect genomic clines, we found stronger correlations for ancestry at epistatically interacting loci than non-epistatically interacting loci (i.e. pairwise associations are evident; results not shown). This result indicates that including ancestry at individual loci as predictors in the Bayesian genomic cline model might facilitate the detection of pairwise epistasis. We have previously used a similar approach in an empirical study of hybridization in mice with the likelihood-based genomic cline model (Teeter *et al.* 2010), and we will explore similar extensions to the Bayesian genomic cline model in the future.

The Bayesian genomic cline models also provides a means to assess geographic variation in the genomic architecture of reproductive isolation. A key question in speciation research is whether the genetics of speciation for a pair of taxa vary spatially and among contexts in which species meet (e.g. Buerkle & Rieseberg 2001; Yanchukov *et al.* 2006; Nolte *et al.* 2009; Teeter *et al.* 2010). Spatial variation in the genetic architecture of speciation could occur because of variation in exogenous selection pressures associated with differences in the abiotic or biotic environment at different regions of contact. Additionally, spatial variation in the genetic architecture of reproductive isolation could result from differences in allele frequencies and segregating variation across the geographic ranges of hybridizing species.

Whereas the genomic cline model has the potential to advance our understanding of the speciation process, it also has a number of limitations. Some of these limitations can be circumvented through refinement of the model, but other limitations are more fundamental to the dynamic complexity of admixture and introgression. The conditional priors we specified for the locus effects on cline parameters (i.e. γ_i and ζ_i) assume that each genomic cline parameter is an independent sample from an underlying genome-level distribution (specifically, a Gaussian distribution with mean zero and precision τ). This may be a reasonable assumption in many circumstances. However, this assumption is likely violated when tightly linked genetic regions are analysed, as the cline parameters describing rates of introgression for linked markers are not independent. A potential solution to this issue is to replace these priors with distributions that account for correlations in rates of introgression among physically linked genetic regions. For example, Guo *et al.* (2009) specified a conditional autoregressive prior for F_{ST} to account for autocorrelations in genetic differentiation along chromosomes. Additionally, the conditional probability of locus-specific ancestry could be adjusted to account for co-inherited blocks of ancestry (i.e. 'admixture linkage disequilibrium') as described by Falush *et al.* (2003). We are currently investigating these and other approaches to account for non-independence of genomic clines for linked markers. A second limitation of the current model is that we assume genotypes are known without error. This assumption is valid or nearly valid for many traditional genetic marker data (e.g. SNPs), but it is violated for next-generation sequence data because finite sequence coverage and errors lead to uncertainty in diploid genotypes (Hellmann *et al.* 2008; Lynch 2009; Gompert *et al.* 2011). Ameliorating this issue is rather trivial and simply requires substituting two functions for Eqn 2, which denotes the probability of the observed genetic data, given the parental allele frequencies and ancestries, $P(\mathbf{x} \mid \boldsymbol{\pi}, \mathbf{z})$. One substitute function would specify the probability of the observed data ($\mathbf{x}$), given the unobserved genotypes ($\mathbf{g}$), and a second function would specify $P(\mathbf{g} \mid \boldsymbol{\pi}, \mathbf{z})$. The latter is equivalent to Eqn 2 with $\mathbf{g}$ substituted for $\mathbf{x}$, whereas the first function would depend on the specific manner the sequence data were generated and what aspects of uncertainty were being modelled (e.g. Hellmann *et al.* 2008; Gompert *et al.* 2011; Gompert & Buerkle 2011).

A more fundamental issue that arises in genome scans generally is what fraction of the genome has experienced selection and contributed to reproductive isolation. In some cases, loci with extreme or 'outlier' genomic cline parameters will not be the only genetic regions that contribute to reproductive isolation. If a large number of genetic regions contribute to reproductive isolation, only those with the strongest association with isolation are likely to be classified as 'outliers', whereas genetic regions with more moderate associations with selection and isolation will account for much of the distribution of cline parameters. Furthermore, although loci with unlikely 'outlier' genomic clines are expected to be enriched for genetic regions affecting fitness of admixed individuals, the 'outliers' also will often include genetic regions not affected by selection. This is because stochastic processes can mimic the effects of selection. Fortunately, analysis of simulated data in this study indicates that stochastic drift is more likely to affect genomic clines by altering α_i (i.e. by causing excess ancestry of one parental population in the admixed population) than by altering β_i (i.e. the rate of transition in ancestry along the admixture gradient). This means that stochastic variation in genomic clines is more likely to mimic the effect of adaptive introgression than reproductive isolation (i.e. underdominance affects α or β, whereas directional selection only influences α). Biased mutation (i.e. biased gene conversion or segregation distortion) could have similar or

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even more severe effects on genomic clines. Obtaining a true neutral distribution to contrast with observed genomic clines would be quite difficult, as it would require simulating neutral introgression given the precise demographic history and mutation landscape experienced by the admixed population, which is often unknown (but see, Fitzpatrick *et al.* 2010). Finally, although we simulated data under a number of demographic histories, we did not explore the effect of complex spatial structure or simulate very recent or very old hybrid zones. The distribution of cline parameters for some demographic histories might be quite different from what we have explored in this study and it is possible that the Bayesian genomic cline model might not perform as well for some specific cases.

Mus *hybrid zones*

The primary finding from our genomic cline analysis of two admixed house mouse populations was a clear pattern of considerable variation in the rate and form of introgression across the genome. The magnitude of variation in genomic cline parameters for these empirical data exceeds that estimated from the simulated data sets. Genomic cline parameter estimates indicate that some genetic regions introgress relatively freely, whereas other regions introgress much less. Many genetic regions also displayed excess *M. domesticus* or *M. musculus* ancestry in the hybrids. Consistent with previous studies of hybridization between *M. domesticus* and *M. musculus* two X-linked markers were among the genetic regions with the lowest rates of introgression (i.e. high positive estimates of β_{ij}; Tucker *et al.* 1992; Payseur *et al.* 2004; Macholan *et al.* 2007; Teeter *et al.* 2008, 2010). One of these markers, X.089Pola1, was also one of three genetic regions classified as 'outliers' based on our analysis. Also consistent with this finding, laboratory crosses in mice and other organisms indicate that hybrid sterility often maps to the X-chromosome (Coyne & Orr 2004; Vyskočilová *et al.* 2005; Good *et al.* 2008). Nonetheless, several autosomal genetic regions showed equally low rates of introgression (e.g. 5.097 in the Bavaria population and 9.052 in Saxony), suggesting the genetic basis of reproductive isolation is not limited to sex chromosomes. Results from previous geographic and likelihood-based genomic cline analysis also indicated low rates of introgression for autosomal genetic region 5.097 (Teeter *et al.* 2008, 2010), consistent with the hypothesis that this genetic region is associated with reproductive isolation. Finally, the number of genetic regions classified as 'outliers' using the Bayesian genomic cline (three) was far fewer than were classified as inconsistent with neutral explanations based on previous likelihood-based genomic cline analysis of these data (35 in Bavaria and 40 in Saxony). Given the lower false positive rate for simulated data using the current model, this likely indicates that the Bayesian model provides a more accurate null distribution for genomic cline parameters (but see caveats in the previous section).

A second key finding of our analysis was that both genetic regions with similar and dissimilar patterns of introgression between the Bavarian and Saxony admixed population were common. Teeter *et al.* (2010) reached the same general conclusion based on previous geographic and genomic cline analysis of these data. For example, consistent with Teeter *et al.* (2010), estimated genomic clines for 1.046 and X.089Pola1 were quite similar between the two admixed populations, whereas estimated genomic clines for 5.007 and X.099Xist differed considerably. However, for some loci, we detected marked differences in Bayesian genomic clines between admixed population that were not detected by the likelihood-based analysis conducted by Teeter *et al.* (2010), such as genetic region 16.014. Regardless of these minor discrepancies, the results of our re-analysis are largely consistent with those reported by Teeter *et al.* (2010) and suggest considerable geographic variation in patterns of introgression among populations. Although this pattern very likely reflects stochastic variation among these populations, it is probable that the variation is enhanced by geographic variation in the genetic architecture of reproductive isolation.

Conclusions

Hybridization in natural populations provides a unique opportunity to study the factors and forces driving adaptation and speciation (Endler 1977; Hewitt 1988; Barton & Hewitt 1989; Arnold 1997; Mallet 2005). Consequently, previous research on hybrid zones has led to fundamental insights into the speciation process (e.g. Barton & Hewitt 1985; Rieseberg *et al.* 1999; Coyne & Orr 2004; Payseur *et al.* 2004; Nolte *et al.* 2009). Further progress in understanding speciation will be facilitated by a robust model framework for quantifying patterns of introgression for individual genetic regions and comparing these patterns across the genome and among populations. The Bayesian genomic cline model we have presented and explored in this study fills this gap. This model builds on extensive theoretical research on geographic clines (Endler 1977; Barton 1983; Barton & Hewitt 1985; Porter *et al.* 1997), as well as previous likelihood-based genomic cline models (Lexer *et al.* 2007; Gompert & Buerkle 2009). The Bayesian genomic cline model is also closely related to widely used models for estimating admixture proportions implemented in *structure* (Pritchard *et al.* 2000), but serves a different

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purpose. Specifically, whereas the *structure* admixture model assumes a constant probability of ancestry across the genome of an individual, the Bayesian genomic cline model allows the probability of ancestry to vary among loci across the genome, with quantification of this variation as one of its primary applications. This aspect of the model makes it ideally suited for dissecting the genetic architecture of reproductive isolation, which is a fundamental component of speciation research.

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The authors are broadly interested in population genetics and the genetics of adaptation and speciation. Z.G. is a Ph.D. student and C.A.B. is an associate professor at the University of Wyoming.

Appendix

MCMC algorithm

We use MCMC to sample the posterior probability distribution specified by $P(\boldsymbol{\pi_0},\boldsymbol{\pi_1},\mathbf{z},\mathbf{h},\gamma,\boldsymbol{\zeta},\boldsymbol{\eta},\boldsymbol{\kappa},\tau_\alpha,\tau_\beta,\nu,\omega \mid \boldsymbol{\psi_0},\boldsymbol{\psi_1},\mathbf{x})$ and obtain estimates of the genomic cline parameters. The full posterior probability distributions for $\boldsymbol{\pi_0}$ and $\boldsymbol{\pi_1}$ are given by Eqn 1. We sample each element of $\mathbf{z}$ directly from its full conditional distribution $p(z_{ijna} \mid \boldsymbol{\pi_{i0}},\boldsymbol{\pi_{i1}},\mathbf{x_{ijna}},\phi_{ijna})$. We update each element of $\mathbf{h}$ using the Metropolis algorithm with a uniform proposal distribution over the interval $[(h_n-0.1), (h_n+0.1)]$. Similarly, we sample each element of the locus effects γ and $\boldsymbol{\zeta}$ using the Metropolis algorithm with proposal values equal to the current parameter value plus a random deviate from a Gaussian distribution with mean zero and a specified variance. For each locus, proposal values for the nested population effects $\boldsymbol{\eta}$ and $\boldsymbol{\kappa}$ for population 0 to population $J-1$ are obtained by adding a random deviate from a multivariate normal distribution to the current value, whereas the value for population J is set to $-1*\sum_{J-1}\Theta_j$, where Θ_j denotes either nested population effect. This algorithm imposes the 'sum-to-zero' constraints using reparameterization by sweeping described by Vines *et al.* (1996). All precision parameters $(\tau_\alpha,\tau_\beta,\nu,\omega)$ are sampled directly from their full conditional distributions.

Supporting information

Additional supporting information may be found in the online version of this article.

Fig. S1 *Mus* genomic clines. Solid lines denote point estimates of genomic clines for the Bavarian (black) and Saxony (grey) admixed populations. 95% ETPI for these cline estimates are denoted with dashed lines. The name of each locus (chromosome number followed by map location) is also given.

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